

Agentic Social Affordance Framework (ASAF): Agent Identity Design as a Collaboration Interface in Multi-Agent Systems

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April 19, 2026 | Preprint | License: CC BY 4.0

Abstract

As AI systems evolve from single conversational agents to complex multi-agent architectures, a critical design dimension has been overlooked: how the social identity of individual agents shapes human behavior within the collaboration. This paper introduces the Agentic Social Affordance Framework (ASAF), a theoretical framework that extends Social Affordance theory into the context of multi-agent AI systems. We propose that agent identity design functions not merely as a user interface convention, but as a collaboration interface—structuring how users perceive, approach, and engage with each agent, and thereby influencing the quality of Human-Agent collaboration outcomes. Specifically, the social affordance layer constitutes an independent design dimension orthogonal to engineering orchestration: the two represent distinct decision spaces that cannot be derived from each other. ASAF comprises three mechanisms: Identity Signaling, Behavioral Priming, and Collaborative Governance, and specifies their boundary conditions through a four-tier Identity Signal Fidelity Spectrum and an individual-difference moderating variable (anthropomorphizing vs. instrumentalizing cognitive style). We situate ASAF in relation to existing affordance theory and the CASA paradigm, delineating where ASAF’s multi-agent, topology-level predictions exceed the explanatory scope of dyadic frameworks. We discuss implications for multi-agent system design and outline directions for future empirical validation, including a factorial design for testing design-space orthogonality.

Keywords: Social Affordance, Multi-Agent Systems, Human-Agent Interaction, Agent Identity Design, Agentic AI, HCI, Human-in-the-Loop, Cognitive Scaffolding

1 Introduction

Recent advancements in multi-agent artificial intelligence have driven increased research on orchestrational efficiency, token optimization, and inter-agent communication protocols (e.g., Hong et al., 2023). While this engineering-centric discourse has successfully established the technical scaffolding for collaborative AI, it inherently marginalizes the human-computer interaction (HCI) dimension. Modern multi-agent systems are rarely closed causal loops; they exist primarily in Human-in-the-Loop (HitL) configurations where human operators must direct, parse, and trust the outputs of highly specialized, distinct node agents.

The impetus for this conceptual framework emerged directly from the author’s industrial deployment challenges rather than isolated laboratory experiments. During the rapid proliferation of orchestration engines (e.g., LangGraph, AutoGen), the developer community overwhelmingly anthropomorphized agent workflows into corporate team structures (detailed in Section 2.3), focusing intensely on which execution engine to adopt. However, when scaling a custom collaborative system from a single conversational assistant to a scalable multi-agent personal productivity architecture comprising dozens of specialized nodes (the *ChronicleCore* system detailed in Section 4), a critical reality became apparent: the true bottleneck is rarely the underlying orchestration framework. Instead, the fundamental friction point is *Context Governance*: the human operator’s capacity to navigate and interact with disparate computational nodes without inducing cognitive collapse or systemic persona drift.

This realization bridges the gap to established literature. Historical meta-analyses in human-robot interaction suggest that human trust and collaborative effectiveness are significantly influenced by an agent’s attribute-based design and perceived role (Hancock et al., 2011). In the context of large language models, this dynamic manifests through “role-play,” where system prompts cast the model into specific functional and social identities, establishing boundaries for its computational behavior (Shanahan et al., 2023). However, the implications of these role-playing dynamics observed within a cohesive multi-agent topology remain critically under-theorized.

This paper argues that the social identity of individual agents within a network functions not merely as a cosmetic persona, but as a structural collaboration interface. Anthropomorphic architecture in multi-agent systems is neither an emotional user-experience feature nor an arbitrary engineering choice; rather, it is an inevitable design response to human cognitive architecture confronting systems whose scale exceeds unassisted working memory. We adopt Cowan’s (2001) working memory limit ($N \approx 4$ items) as an approximate problem-scale threshold. While Cowan’s measure concerns simultaneous item retention in short-term memory rather than active cross-session navigation, it establishes the cognitive order of magnitude at which unstructured information management becomes untenable; the cognitive offloading literature (Risko & Gilbert, 2016) demonstrates that when task demands approach such capacity limits, users spontaneously seek external scaffolding structures. ASAF posits that at this approximate scale ($N \geq 4$ unstructured functional profiles), the application of social schemas (chunking) transforms from a superficial preference into a structural necessity (i.e., an indispensable design consideration, not a guaranteed effect for all users; effect magnitude remains moderated by individual cognitive style, as specified in H1).

ASAF distinguishes itself from existing theories (such as CASA, classical role theory, and distributed cognition) through a specific structural claim: the social affordance layer constitutes an *independent design dimension* orthogonal to engineering orchestration. That is, designing the social identity layer and designing the engineering orchestration layer are distinct decision spaces—one cannot be derived from the other. This is a claim about the structure of the design problem, not about effect-independence: because Social Affordances are co-constructed between system design and user perception (Section 3.1), their realized effects remain moderated by individual cognitive style, as formalized in H1’s moderating variable.

To clarify the boundary with CASA (Reeves & Nass, 1996): CASA’s core prediction—that humans automatically apply social norms to computers—can account for the basic Identity Signaling effect at the dyadic level (i.e., a single user responding socially to a single personified agent). However, CASA is structurally a dyadic theory; it does not generate predictions about (a) the scaling threshold at which social identity shifts from optional to structurally necessary ($N \geq 4$; H1), (b) the differential oversight calibration across agents occupying distinct topological roles within a system (H3), or (c) the orthogonality of the social design space to the engineering design space. No extension of CASA—including aggregated or conformity-based variants (cf. Song et al., 2025)—generates the topological role differentiation prediction in H3, because CASA lacks a structural vocabulary for agent-topology-level oversight calibration. ASAF’s independent contribution lies precisely in these multi-agent, topology-level, and design-theoretic dimensions that fall outside CASA’s explanatory scope.

While existing theories explain isolated phenomena (e.g., dyadic trust or general cognitive off-loading), ASAF predicts that the social affordance design space governs the overall interaction legibility of the system through a design axis that is irreducible to its backend architecture. Under this framework, an engineering-fragile architecture can still be rendered operable by leveraging strong social schemas, whereas an engineering-robust system stripped of legible social affordances will suffer from severe interactional friction at scale. This design-space orthogonality is supported by design-layer analysis: engineering restructuring (e.g., changing orchestration frameworks or inter-agent protocols) does not entail any particular social identity configuration, and conversely, redesigning agent personas does not alter the engineering topology. H1 (Section 5) provides a complementary empirical test of whether social identity design carries independent explanatory power beyond functional labeling, though a full test of orthogonality would require a factorial design that simultaneously manipulates both axes (see Section 7). Consequently, treating multi-agent design purely as a software engineering decomposition problem risks creating structurally coherent but interactionally unreadable systems.

This paper makes the following contributions:

1. We introduce the Agentic Social Affordance Framework (ASAF), distinguishing its predictive power by positioning anthropomorphic identity not as an empathetic illusion, but as a necessary cognitive load-reduction interface for human operators managing systems that exceed working memory capacity ($N \geq 4$).
2. We identify and define three core mechanisms through which ASAF operates: Identity Signaling, Behavioral Priming, and Collaborative Governance.

3. We situate ASAF in relation to existing affordance theory (Gibson, 1979; Norman, 1988) and automation reliance literature (Parasuraman & Riley, 1997; Lee & See, 2004).
 4. We provide operational design boundaries, formally defining how individual differences in cognitive style (anthropomorphizing vs. instrumentalizing) moderate the framework’s predictive validity.
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2 Background and Related Work

2.1 Affordance Theory: From Gibson to Social Affordance

The concept of affordance was introduced by Gibson (1979) as a relational property between an actor and their environment: the action possibilities that an environment offers to a particular actor. Affordances are neither purely objective properties of objects nor purely subjective constructions of the mind; they emerge from the relationship between the two.

Norman (1988; 1999) operationalized affordance for design contexts, distinguishing between real affordances (what an object actually allows) and perceived affordances (what users believe it allows). Norman’s contribution established affordance as a central concept in human-computer interaction, directing attention to how design choices shape user perception and behavior.

Social Affordance extends this framework to the social dimensions of technology. Where Gibson’s affordances describe physical action possibilities and Norman’s describe functional ones, Social Affordances describe the social interaction possibilities that a system’s design signals to its users (Hutchby, 2001). During the Web 2.0 era, this was extensively applied to analyze how the spatial architecture of social network sites shaped “networked publics” and structured human connection (Boyd, 2010). Nagy and Neff (2015) advanced this domain by introducing the concept of *imagined affordances*, arguing that a user’s *belief* about what a technology or algorithmic system can do is just as determinant of their behavior as the technology’s hard-coded capabilities. This transitioned affordance theory from analyzing static interfaces to analyzing socially constructed expectations.

Recent sociological work has begun bridging this framework into artificial intelligence. A critical conceptual precursor is Sundar’s (2020) framework for Human-AI Interaction, which builds on his earlier Theory of Interactive Media Effects (TIME) and examines the tension between human and machine agency. Sundar identifies the “machine heuristic”—where users apply mental shortcuts to evaluate AI outputs based on the mere presence of a machine interface. While Sundar’s machine heuristic operates as a binary trigger (machine vs. human), Identity Signaling in multi-agent systems operates as a multi-valued, role-differentiated signal: users do not merely respond to “a machine” but calibrate their behavior to *which kind of social role* the machine presents. This shift from binary agency detection to structured role-based calibration is precisely the granularity gap that ASAF addresses. Empirical studies on ChatGPT usage in specific cultural contexts illustrate how perceived affordances in LLMs are shaped by user interpretation practices rather than fixed machine properties alone (Haqu

et al., 2025). Anthropomorphic design cues further trigger parasocial relationship dynamics, modifying user behavior without changing the underlying model parameters (Reeves & Nass, 1996; Epley et al., 2007). Together, these findings indicate that affordance theory must evolve past simple “machine heuristics” toward interpreting the explicit social scripts embedded in system design.

2.2 The Chatbot Era: Single-Agent Social Affordance

In the Chatbot era, Social Affordance operated through a single, often ambiguous social identity. ChatGPT, Claude, and similar systems present a unified interface that affords many things—assistance, conversation, analysis, creativity—but commits to none. This ambiguity is a design choice that maximizes perceived utility but minimizes Social Affordance specificity. Users engage with these systems through general-purpose interaction patterns, rarely adjusting their behavior based on a perceived social role of the system.

The limitation is structural: with a single agent identity, Social Affordance can only operate at the level of the whole system. There is no mechanism for users to adjust their interaction style based on the specific capability or social role of a particular component.

2.3 The Evolution of Agentic Roles: The “Virtual Team” Wave and Its Backlash

Multi-agent AI systems introduce a structurally distinct architecture, replacing single conversational models with networks of specialized agents. Beginning in late 2025 and accelerating through the first quarter of 2026, a rapid resurgence of anthropomorphic multi-agent architectures emerged, what we term here the **Virtual Team Wave**. Developers returned to structuring agents as corporate role hierarchies, and in some cases, historical administrative metaphors. Three representative open-source implementations collectively accumulated **169,732 GitHub stars** within this window, surfacing an apparent industry-wide appetite for persona-structured agent fleets.¹

Table 1. Representative open-source multi-agent implementations adopting anthropomorphic role structures (metrics retrieved via GitHub REST API, 2026-04-16).

These implementations illustrate both the scale of the trend and the cross-cultural nature of the underlying design impulse. *The Agency* (Sitarzewski, 2025), which seeded the wave in October 2025, organizes 147 agents into corporate-style divisions (engineering, design, marketing, product), each agent written as “a specialized expert with personality, processes, and proven deliverables.” *gstack* (Tan, 2026), authored by the President of Y Combinator and released in March 2026, operationalizes a single builder’s team as “23 opinionated

¹Repository metrics retrieved via the GitHub REST API (GET /repos/{owner}/{repo}) on 2026-04-16. Aggregate at retrieval: 169,732 stars, 24,954 forks across the three repositories. These figures aggregate three distinct repositories; actual unique observer count is lower due to overlap among early-adopter communities who star multiple related projects—a sampling bias inherent to GitHub stars as a social signal inflated by high-profile distribution events (e.g., curator endorsements). The metric is cited as a directional indicator of industry interest, not as a precise adoption measure.

Table 1: Representative open-source multi-agent implementations adopting anthropomorphic role structures (metrics retrieved via GitHub REST API, 2026-04-16).

Repository	Author	Created	Stars	Forks	Design Metaphor
agency-agents	Sitarzewski	2025-10	80,818	12,937	Corporate divisions
edict	cft0808	2026-02	15,204	1,596	Tang Dynasty governance
gstack	Tan	2026-03	73,710	10,421	Startup builder team
Total			169,732	24,954	

tools that serve as CEO, Designer, Eng Manager, Release Manager, Doc Engineer, and QA,” accumulating over 73,000 stars within its first five weeks. *Edict* (cft0808, 2026; the maintainer publishes under this GitHub handle) departs from the Silicon Valley corporate metaphor entirely, adopting the **Three Departments and Six Ministries** (三省六部) system of Tang Dynasty governance (ca. 618 CE). It distributes agent roles across the Secretariat, the Chancellery, and the Department of State Affairs, operating under a strict inter-agent permission matrix. That the same underlying design impulse surfaces in both Silicon Valley startup vernacular and 1,400-year-old administrative structure suggests that this anthropomorphization is not a local stylistic preference but a recurring cognitive response to the orchestration demands of multi-agent systems. At the commercial frontier, xAI’s Grok 4.20 (released February 2026) deploys four named specialist agents as its default inference architecture (xAI, 2026).² This structural shift towards roleplaying clusters is not an isolated design anomaly but an emergent standard practice. Qualitative studies of early adopters confirm that practitioners organically conceptualize and interact with complex LLM pipelines as “teams” of specialized, task-based collaborators rather than mere software threads (Naik et al., 2025).

However, while Naik et al. (2025) establish *that* early adopters default to this team-oriented mental model, the underlying mechanism shaping interaction remains untheorized. Empirical work concurrently demonstrates that groups of AI agents can function as cohesive social groups, exerting normative social pressure on users through mechanisms analogous to human group dynamics (Song et al., 2025).

Recent empirical work has demonstrated that specific agent properties (e.g., group presence (Song et al., 2025), verbalized uncertainty (Xu et al., 2025), and confidence calibration (Li et al., 2024)) exert measurable influence on human collaboration behavior. However, these findings have yet to be integrated under a unifying design-level framework. ASAF offers such an integration by treating each of these properties as instances of Social Affordance signaling.

However, within the same window a fierce **Anti-Roleplay Backlash** emerged in parallel. The developer community identified a fatal architectural flaw in the “virtual team” model: *information decay during transmission*. Unlike humans who compensate for communication

²xAI’s Grok 4.20 (released February 17, 2026) markets “four expert-mode AI agents” as a core feature of its paid SuperGrok tier (grok.com). The individual agent names—including Benjamin (mathematical reasoning) and Lucas (contrarian analysis)—are observable within the paid product interface (SuperGrok subscription required).

gaps with shared cultural context, LLMs passing synthesized documents down a rigid assembly line suffer from continuous context loss and “reasoning drift.” Artificially imposing strict role boundaries restricts the model’s inherent cross-domain capabilities, creating false walls precisely where valuable edge-case reasoning tends to occur.

As a result, engineering consensus has swung violently away from persona-driven pipelines. Leading infrastructure providers now explicitly warn against fragmented agent architectures (e.g., Cognition AI’s critique of multi-agent fragility; Yan, 2025), prioritizing pure deterministic paradigms instead. Frontier labs such as Anthropic (2025a), for instance, have publicly advocated “Context Engineering” and the use of explicit shared state files (e.g., persistent `progress.txt` or Git-based logs) over sequential roleplay. In this current paradigm, agent identity is widely dismissed as a dangerous, anthropomorphic illusion.

ASAF’s mechanisms presuppose that agent identity signals remain perceivable to the human operator across interactions. To contextualize ASAF’s scope against engineering critiques of context decay (e.g., Yan, 2025), we structure agent identity persistence along a four-tier design spectrum:

1. **Pure Prompt Injection:** Personas are defined per-session with no persistence, rendering them highly susceptible to reasoning drift.
2. **Prompt with Persistent Memory:** Personas span sessions via memory access, but lack structured cross-agent auditing.
3. **Structured Identity Enforcement:** Agents possess persistent structural modules, hard-gated orchestration layers, and cross-agent validation mechanisms (e.g., the ChronicleCore architecture detailed in Section 4).
4. **Model-Driven Routing:** Specialist routing occurs natively at inference-time and cannot be bypassed by the user (e.g., Grok 4.20).

ASAF scopes its strongest predictions to systems operating at Tier 3 and 4, where Social Affordances are structurally enforced. The framework’s predictions apply partially to Tier 2, while Tier 1 serves as a boundary condition where framework effectiveness degrades proportionally with identity signal fidelity. Yan’s (2025) critique accurately describes the structural fragility inherent to Tier 1 and 2 deployments. However, his critique targets the engineering layer (context fragility), while ASAF operates on the cognitive layer (how human operators navigate deployed architecture). These are orthogonal concerns: a structurally fragile system can still benefit from Social Affordance design, and a structurally robust system without it remains interactionally illegible. The appropriate tier for any deployment is determined by operational requirements and risk tolerance, not by technological epoch. ASAF’s predictive validity scales with identity signal fidelity across this spectrum—its mechanisms remain conceptually applicable at any tier, with strongest empirical predictions at Tier 3-4 where structural enforcement removes identity drift as a confound.

2.4 The Cognitive Interface Imperative

Building upon early theories of “cognitive technologies” that reorganize human mental functioning (Pea, 1985; Salomon, 1993), and on the growing literature on cognitive offloading in everyday tool use (Risko & Gilbert, 2016), we conceptually define a well-designed multi-

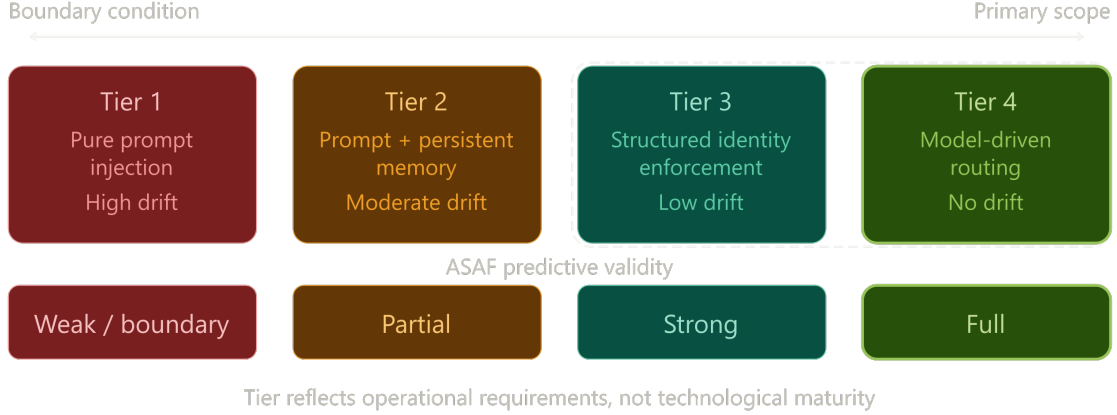


Figure 1: The Identity Signal Fidelity Spectrum. ASAF’s predictive validity scales with the structural enforcement of agent identity signals. Tier 1 (ephemeral prompt injection) represents a boundary condition where identity drift degrades framework predictions proportionally. Tiers 3–4, where identity signals are structurally enforced or model-native, constitute the primary scope of ASAF’s strongest empirical predictions. The spectrum is not a developmental trajectory—deployment tier is determined by operational requirements, not technological maturity.

agent fleet as an external cognitive scaffold. A system whose structure mirrors how the human architect’s brain naturally decomposes problems externalizes operations that would otherwise exhaust working memory. When each agent carries a legible social identity, this scaffolding effect becomes structured and predictable rather than accidental (Amershi et al., 2019). The potential for deliberate Human-Agent cognitive synergy of this kind remains largely absent from current orchestration literature.

Reducing multi-agent architectures to purely functional software pipelines ignores the true operational bottleneck of Human-in-the-Loop systems. The ultimate hardware ceiling of any highly scaled agentic system is not GPU capacity or API rate limits; it is the *human operator’s own cognitive architecture*. If an agent is treated purely as a software tool, the interaction remains strictly dictated by the user’s working memory and procedural management limits.

Foundational HCI research is unambiguous here: effective system design must align with human mental models, not just maximize output accuracy (Bansal et al., 2019). In the context of LLMs, Shanahan et al. (2023) frame these models as “simulators of text-generating processes” capable of instantiating diverse “simulacra” (personas or roles). Agent Identity, therefore, acts as an **Interpretability Prior**: by casting the underlying simulator into a distinct social role, a developer constrains behavioral unpredictability while simultaneously giving the human operator a familiar heuristic for predicting the agent’s “error boundaries” and epistemic scope.

A critical boundary condition of ASAF is its explicit dependence on individual differences in cognitive style as a formal moderating variable. Specifically, the tendency to *anthropomorphize* versus *instrumentalize* computational tools (Epley et al., 2007) moderates the magnitude of

the framework’s predicted effects. ASAF remains theoretically applicable across the entire user spectrum, but the effect size of identity design on cognitive offloading is directly proportional to a user’s natural propensity to employ social scripts. For users who adopt a strictly instrumentalist stance (actively resisting social script activation and treating agents purely as mechanistic software threads), the cognitive scaffolding effect is significantly attenuated. For these instrumentalizing users, encountering role-based design at scale ($N \geq 4$) may introduce cognitive overhead rather than reduce it, as they force themselves to track raw functional boundaries rather than relying on intuitive social heuristics. By positioning anthropomorphic tendency as a moderating variable rather than an absolute structural prerequisite, ASAF generates distinct, falsifiable interaction trajectories for mixed user populations.

3 The Agentic Social Affordance Framework (ASAF)

ASAF proposes that in multi-agent AI systems with Human-in-the-Loop configurations, agent identity design functions as a Social Affordance layer. Before detailing its operational mechanisms, it is critical to anchor this framework in its sociological and cognitive psychology origins.

3.1 Theoretical Foundations: Cognition, Sociology, and Co-construction

In the context of ASAF, a *Social Affordance* is operationally defined as a relational property of an agent’s designed identity that signals to a human user the interaction possibilities, epistemic scope, and appropriate collaboration modality specific to that agent’s declared role. Unlike physical affordances (Gibson, 1979) or functional affordances (Norman, 1988), Social Affordances in multi-agent systems are *co-constructed* between the system’s design and the user’s perception (Haquu et al., 2025). The affordance emerges because the human user brings their own pre-existing cultural scripts to interpreting the agent’s persona. Consequently, ASAF’s claim that the social affordance layer constitutes an independent design dimension (Section 1) refers to design-space orthogonality—the social layer and the engineering layer represent distinct axes of design decisions—not to effect-independence. The realized magnitude of any Social Affordance is inherently user-moderated, scaling with the individual’s propensity to activate social scripts (see H1, Section 5).

Therefore, the efficacy of ASAF does not depend on the LLM possessing actual sociological awareness, but on the human operator possessing ingrained social heuristic scripts. We anchor this claim in three distinct theoretical pillars:

First, sociological **Role Theory** (Biddle, 1986) establishes that societal roles carry specific “role expectations.” ASAF posits that when an AI system assumes a clear persona, it taps into these deeply ingrained human cognitive expectations, implicitly nudging the human user to align their prompt to the expected interaction space without necessitating exhaustive explicit instructions.

Second, Erving Goffman’s **Dramaturgical Analysis** (1959) argues that social interaction is a theatrical performance. In LLM contexts, the system prompt and context window serve as the “back stage” (the hidden mechanisms), while the agent’s identity serves as the “front stage.” This front stage facilitates a defined “definition of the situation,” allowing the human operator to construct meaning based on familiar cultural expectations rather than deciphering raw mechanistic outputs.

Finally, cognitive psychology’s **Distributed Cognition** (Hutchins, 1995) models how cognitive load is offloaded into external environments and societal structures. A multi-agent framework categorizes complex computational operations into discrete social identities. ASAF hypothesizes that this enables human operators to intelligently route complex tasks by querying familiar social abstractions (e.g., “The Critic,” “The Explorer”) rather than expending working memory to track unlabelled, raw computational states.

These three foundational theories manifest in ASAF through three distinct but conceptually interdependent mechanisms, operating across three levels of analysis. Identity Signaling functions at the cognitive level, shaping how an individual user approaches a single agent before any interaction occurs. Behavioral Priming functions at the interactional level, describing how that initial framing propagates into sustained changes in user input behavior across the session. Collaborative Governance functions at the system topology level, operating as a bridging concept across two temporal dimensions. At *design-time*, establishing deliberate identity structures (e.g., configuring adversarial roles) serves as the architectural prerequisite that enables distinct Social Affordances to exist. At *run-time*, this topology governs the emergent dynamics—specifically, how human operators differentially calibrate their oversight based on the cascaded effects of Identity Signaling and Behavioral Priming. This structural interplay establishes the human cognitive interface in multi-agent systems as an independent design dimension, standing alongside the engineering orchestration layer rather than subordinate to it.

3.2 Mechanism 1: Identity Signaling

Definition: Agent identity design activates social scripts in users before interaction begins, reducing cognitive friction and establishing structural interaction expectations.

Recent scholarship in HCI highlights the growing necessity of “thoughtful persona design” in LLMs, predominantly focusing on elements like voice, empathy, and emotional demographics to enhance single-agent interaction quality (Zargham et al., 2024). ASAF diverges from this paradigm. In multi-agent contexts, Identity Signaling is stripped of its primary function to forge “emotional empathy” or “rapport.” Instead, identity is operationalized purely as a structural boundary marker.

ASAF posits that when a user encounters an agent with a defined identity (for instance, an architecture deploying “Benjamin, specializing in mathematical deduction” alongside “Lucas, a dedicated contrarian analyst”), they do not begin interaction from a blank slate. We theorize that they bring to bear existing social scripts for interacting with people who occupy similar analogous roles. These scripts implicitly afford a sense of what kind of input is appropriate, what level of formality is expected, and what type of output to anticipate.

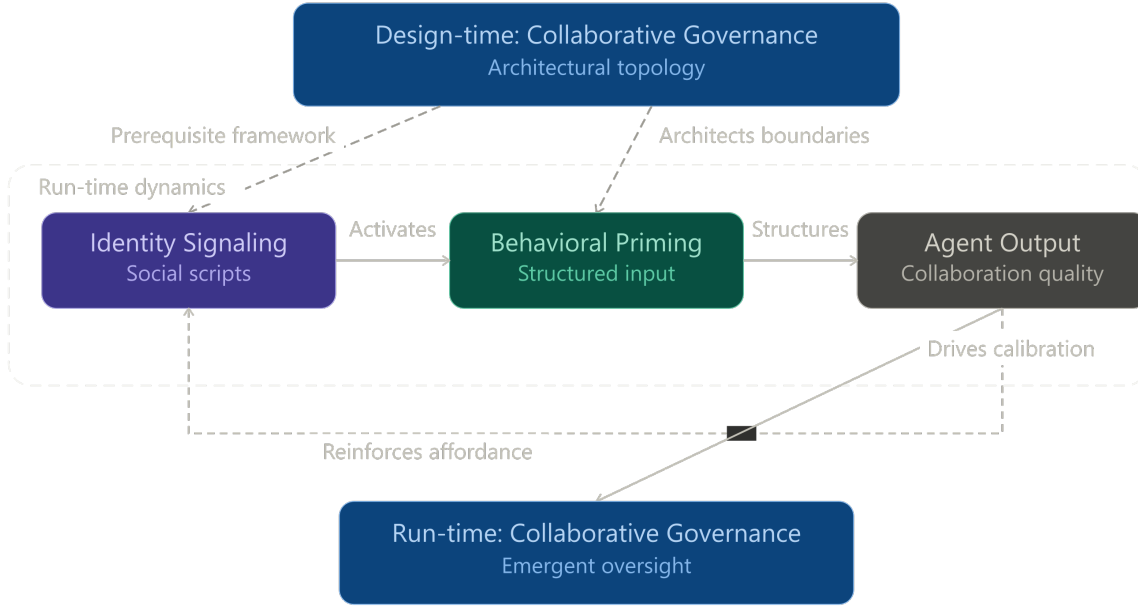


Figure 2: The dual dimensions of Collaborative Governance. Design-time topology acts as the architectural prerequisite for interaction, while run-time emergent governance reflects the human operator’s differential oversight.

This mechanism predicts that social identity will outperform functionally equivalent but non-personified labels (H1: condition a > b) for three specific reasons. First, *activation automaticity*: social role scripts are acquired through a lifetime of interpersonal interaction and activate involuntarily upon encounter (Epley et al., 2007), whereas functional descriptions (e.g., “logical analysis module”) require deliberate parsing to derive appropriate interaction norms. Second, *default coverage breadth*: a social role implicitly encodes not only capability boundaries but also tone expectations, formality level, error tolerance, and appropriate challenge behavior—behavioral norms that a functional label does not carry. Third, *normative load*: social roles carry implicit behavioral obligations for *both parties* in the interaction (Biddle, 1986); a user interacting with a “critic” feels normatively compelled to present defensible arguments, an effect that a label reading “verification subsystem” does not generate. These three properties collectively predict that social identity signals will produce richer behavioral adaptation than equivalent functional information alone.

This pre-interaction activation is the core of Identity Signaling. The Social Affordance operates at the moment of encounter, structuring the user’s approach to the interaction and strategically reducing the cognitive effort required to determine “how should I context-prompt this system?” This creates a specific scaling advantage in multi-agent environments. When managing four or more agents ($N \geq 4$), which is the theoretical ceiling for human chunking of unstructured functional profiles (Cowan, 2001), agent identities sidestep working memory limits by engaging pre-existing social schemas (Epley et al., 2007) rather than requiring the operator to manually track abstract architectural boundaries.

This pre-interaction activation also resolves ambiguities in how users perceive agent output parameters. Consider the counterintuitive finding by Xu et al. (2025), who demonstrated that expressing medium uncertainty (e.g., “I think this might be...”) frequently outperforms absolute confidence in human-AI teaming. Seen through the lens of Identity Signaling, this makes complete sense. We do not cognitively expect an “ideator” to offer design propositions with clinical precision or formal rigor. When such an identity signals uncertainty, the user interprets it not as a system failure, but as a culturally legible invitation to step in and exercise human judgment.

3.3 Mechanism 2: Behavioral Priming

Definition: Agent social identity is anticipated to systematically improve the structural quality of user inputs, creating a positive feedback loop that elevates collaboration outcomes.

Jahani et al. (2024) document how users iteratively learn to adapt their prompting behaviors over time, with significant performance gains emerging from this acquired interaction fluency rather than from model improvements alone.

ASAF’s Behavioral Priming mechanism theorizes how agent identity drives this influence. We postulate that when interacting with what they tacitly treat as a domain-competent counterpart (cf. Reeves & Nass, 1996), users are primed to voluntarily: * Provide more structured, context-rich inputs aligned directly with the agent’s declared domain. * Proactively include constraints, assumptions, and boundary conditions. * Formulate questions at a higher level of structural specificity.

Jahani et al. (2024) document that this fluency typically develops through iterative trial-and-error across multiple sessions. ASAF hypothesizes that strong Identity Signaling may compress this learning curve substantially, activating role-appropriate prompting patterns from the very first interaction.

A secondary, yet crucial implication of Behavioral Priming relates to cognitive bias mitigation. It is well-documented that human users are dangerously susceptible to AI overconfidence, often failing to detect when a model is hallucinating assertively (Li et al., 2024). However, what happens if we deliberately assign the system a socially cautious or adversarial persona? If a user explicitly knows they are interacting with an “auditor” (by definition, a role whose social script foregrounds challenge and verification over generative output), this alone structurally intercepts their default vulnerability to over-reliance. The social frame does the work; no explicit instruction to “be skeptical” is required.

3.4 Mechanism 3: Collaborative Governance

Definition: In Human-in-the-Loop configurations, agent social identity structures the location and mode of human intervention, transforming HitL from a mere safety checkpoint into a collaboration design element.

The conventional framing of Human-in-the-Loop in AI systems is safety-centric: human oversight exists purely to catch errors. This framing is correct but incomplete. The foundational

literature on automation reliance (Parasuraman & Riley, 1997) and the dynamics of human trust in automated systems (Lee & See, 2004) establishes that human operators constantly modulate their reliance and oversight based on the system’s perceived purpose and past performance. ASAF extends this to the multi-agent context, positing that when a system possesses defined agent identities, users do not intervene generically; their oversight calibration is dynamically structured by each agent’s specific social role. In a recent scoping review of 134 top-tier HCI papers, Zhang et al. (2025) constructed an integrated theoretical map of how “agency” and control are distributed in human-AI co-creation. ASAF’s Collaborative Governance mechanism serves to operationalize the specific control dynamics mapped by Zhang. We argue that assigning a social identity to an agent is a specific method of *agency negotiation*. A user understanding that one agent is an “ideator” and another is an “auditor” will trust the ideator’s divergence while heavily scrutinizing the auditor’s final checks, thereby calibrating how deeply to intervene at each stage of the workflow.

This mechanism extends into group dynamics. Empirical research validates that groups of AI agents can function as social groups, exerting “normative social pressure” on human users, particularly through conformity effects when multiple agents agree (Song et al., 2025). Under ASAF’s Collaborative Governance, this pressure can be architecturally countered. By deliberately designing interacting agents with competing identities (e.g., assigning a “Devil’s Advocate” identity to explicitly challenge a “Proposer” identity), developers can artificially break the unanimity that causes dangerous conformity effects, structurally forcing the human operator to exercise critical judgment. The governance structure of the system becomes legible and manipulable through the Social Affordance layer. What was once invisible coordination logic is now a perceivable, interactable interface that the human operator can navigate directly.

4 System Implementation: Illustrative Design Case

Rather than presenting a formal empirical study, we offer the development of *ChronicleCore*—an evolving, single-operator Multi-Agent System—as an **illustrative design case** and a design rationale reconstruction. The researcher simultaneously occupied the roles of system designer, primary operator, and observing analyst. This approach captures the tacit operational knowledge and longitudinal cognitive shifts that structured external observation cannot readily surface, grounding the theoretical permutations of ASAF in sustained, first-person cognitive governance.

This paper was itself produced under a multi-agent AI collaboration configuration (see Acknowledgments for specific tools and versions). All conceptual contributions (the ASAF framework, its mechanisms, and hypotheses) originated entirely from the human author and all cited works were manually verified prior to inclusion. The author acknowledges the circular evidentiary structure this creates—using a multi-agent configuration to argue for the value of multi-agent configurations. This methodological reflexivity is part of the framework’s boundary conditions rather than its validation.

To illustrate how ASAF’s mechanisms manifest in a deployed environment, we present a reflective case study of *ChronicleCore*, an evolving, single-operator Multi-Agent System. At the time of writing (April 2026), the architecture comprises 38 active Human-in-the-Loop expert nodes, designed to organically accommodate incremental agent addition as operational needs dictate. Operated by the author as a personal productivity architecture, this case serves as a design rationale grounded in sustained first-person cognitive governance. While Park et al. (2023) famously demonstrated the potential of multi-agent societies to *simulate* human societal behavior, ChronicleCore applies a conceptual inversion: it utilizes deliberately designed societal structures (agent identities) to *constrain and orchestrate* the reasoning pathways of a single human operator.

4.1 Decoupling Agents by Epistemic Role

When a single operator scales a system to several dozen specialized agents, the primary challenge transcends tool orchestration and enters the realm of personal cognitive governance. A single, monolithic chatbot affords generic conversational prompting, but a highly populated agent network risks cognitive overload for the human operator and “persona drift” among the agents themselves. To counteract this, ChronicleCore physically decouples the system into distinct operational pillars, each functioning as a highly specific Social Affordance boundary.

Rather than deploying agents as indistinguishable utility nodes, the architecture categorizes them by their epistemic and rhetorical constraints. The system separates strategic routing nodes (which allocate global context but possess no execution capabilities) from sensory nodes responsible purely for data ingestion. It isolates aesthetic alignment tasks from adversarial verification layers. By granting each agent localized context paired with a heavily defined social identity (e.g., naming a verification node the “Inquisitor” (see Appendix A)), the system is designed to structurally force the human operator to calibrate their oversight. This architectural decoupling was designed precisely to shape the operator’s interaction patterns. By structurally enforcing the separation—where an adversarial node is incapable of executing generalized code generation and is restricted exclusively to probing logical vulnerabilities—the system preempts persona drift. The design enforces the boundary, ensuring that the operator cannot bypass the intended Social Affordance without breaking the system’s operational protocol.

4.2 Sustaining Identity Boundaries: A Design Perspective

The persistence challenge in a multi-agent system with differentiated social identities is not only intra-agent but inter-agent: ensuring that Agent A’s epistemic and rhetorical profile does not assimilate toward Agent B’s over repeated interactions. ChronicleCore’s design addresses this through two operational practices.

Memory Crystallization separates transient reasoning logs from identity-weighted constraint rules encoded in each agent’s persistent SKILL.md layer; only the latter are promoted into long-term state, shielding the agent’s social identity from accumulated task-specific reasoning drift. *Personality Variance Audit* is a systemic cross-agent inspection—operationalized in ChronicleCore through the Inquisitor node’s VETO threshold (Iron Law 5; see Appendix

A, §A.3)—that detects rhetorical and epistemic convergence between agents intended to maintain distinct social identities. Whereas the Personality Uniqueness Check (Section 6) operates at design-time to prevent redundant social affordance signals when new agents are introduced, the Personality Variance Audit operates continuously at runtime to detect epistemic convergence between already-deployed agents.

These are design patterns for applying existing memory mechanisms in service of Social Affordance integrity. Sustained operation supports the postulate that sharply defined, non-overlapping identities facilitate distinct human interaction patterns, while blurred identities give way to operational genericism and diminished collaboration quality.

5 Formalizing ASAF: Testable Hypotheses

To transition ASAF from a descriptive framework into an empirically verifiable model, we formulate three core hypotheses aligned with Human-Computer Interaction (HCI) methodologies. These hypotheses explicitly address potential confounds, such as individual baseline tendencies (anthropomorphizing vs. instrumentalizing) and alternative explanations (the novelty effect), ensuring the framework’s falsifiability:

- **H1 (Identity Signaling and Interaction Framing):** Users interacting with agents bearing differentiated role identities will produce first-turn prompts exhibiting greater domain alignment than users interacting with generic labels. To discriminate between the effects of social identity and mere task decomposition clarity, validation should employ a three-condition design: (a) *identity-differentiated* agents (persona, name, tone), (b) *functionally labeled* agents (capability descriptions without personification), and (c) *generic* agents (no label). ASAF predicts condition (a) > (b) > (c); if the effect derives solely from task decomposition clarity, conditions (a) and (b) should not differ significantly. This contrast tests the independent explanatory power of social identity design beyond functional labeling; it does not directly test design-space orthogonality, which would require simultaneously manipulating the engineering layer (see Section 7). Furthermore, at or beyond the multi-agent scaling threshold ($N \geq 4$), this effect is significantly moderated by baseline tendencies: users with high anthropomorphizing tendencies will leverage social identities to route tasks with significantly lower perceived cognitive load, whereas strict instrumentalizing users will experience rapid cognitive degradation as they attempt to manually track unstructured parameters.
- **H2 (Behavioral Priming vs. Novelty Effect):** Over a sustained task session, the proportion of structured inputs (e.g., constraint density, explicit boundary conditions) will be significantly higher in identity-differentiated conditions than in generic-label conditions. To distinguish this from a mere **novelty effect**, these behavioral adaptations will not decay over time. Instead, adherence to the agent’s social script will exhibit a dose-response relationship with the intensity of the identity signal (e.g., consistency of tone, boundary reinforcement) and will stabilize over repeated longitudinal interactions.
- **H3 (Collaborative Governance):** In Human-in-the-Loop multi-agent architectures, human intervention patterns will significantly differ across varying agent topologies

predicated on Social Affordance cues. Specifically, users will exhibit higher unverified acceptance of generative outputs from designated “Ideator” identities, while applying significantly higher scrutiny to outputs from assigned “Verifier/Auditor” identities, demonstrating deliberate oversight calibration. Critically, this calibration pattern operates at the level of the system’s topological role configuration—users modulating their intervention depth in response to the overall agent topology, not merely reacting to individual agent roles in isolation—a systemic property that CASA’s dyadic framework cannot structurally generate.

6 Implications for Multi-Agent System Design

ASAF suggests several design principles for practitioners building multi-agent systems with human interaction layers:

- **Design agent identity for the interaction, not the implementation.** The social identity of an agent should be designed from the perspective of what interaction pattern it should elicit from users, not from the perspective of what the underlying model can do. Implementation capabilities constrain the design space; Social Affordance considerations should guide the design choices within that space.
 - **Differentiate identities meaningfully.** If two agents in a system have social identities that elicit similar interaction patterns from users, they are functionally equivalent from a Social Affordance perspective—regardless of their technical differences. Meaningful differentiation requires that each agent’s identity activates a distinct social script. This is analogous to what we term a Personality Uniqueness Check: agents whose behavioral and epistemic profiles are insufficiently distinct provide redundant Social Affordance signals.
 - **Treat HitL configuration as Social Affordance design.** The placement and nature of human intervention points in a multi-agent system should be designed in relation to agent identities, not independently of them. Human oversight is most effective when users understand, through Social Affordance signals, what each agent is responsible for and what kind of human judgment is most relevant at each intervention point.
 - **Separate the social layer from the orchestration layer.** Agent identity design (the social layer) and agent orchestration design (the engineering layer) address different problems and should be designed with different considerations. Conflating them—designing agent roles purely based on engineering decomposition—risks producing social identities that are technically coherent but interaction-incoherent.
-

7 Limitations and Future Research

While ASAF is theoretically grounded and supported by observational evidence from production systems like ChronicleCore, its mechanisms have not yet been subjected to controlled,

large-scale empirical validation. Several limitations follow from this:

- **Measurement challenges.** Social Affordance effects are inherently difficult to isolate and quantify. The influence of agent identity on user behavior is confounded by user expertise, task complexity, system context, and individual differences in social script activation. Future research should develop measurement instruments adapted from existing Social Affordance and Human-AI interaction scales.
- **Context dependency.** ASAF’s mechanisms are likely to vary in salience across different deployment contexts. Enterprise users with high AI literacy may exhibit different Social Affordance responses than general consumers. Task domains with strong professional role norms (legal, medical, financial) may show stronger Identity Signaling effects than open-ended creative tasks.
- **The anthropomorphism boundary.** ASAF deliberately engages with anthropomorphic design elements in agent identity. The literature on human-AI interaction has documented both the benefits and risks of anthropomorphism, including parasocial relationship formation and the potential for contextually inappropriate overtrust in faulty systems (Hancock et al., 2011; Salem et al., 2015). Future work should examine where ASAF’s mechanisms produce beneficial collaboration effects and where they risk crossing into problematic anthropomorphism, establishing empirical boundaries against dangerous over-reliance.
- **Misleading identity signals.** ASAF assumes alignment between the social script activated by an agent’s identity and the agent’s actual architectural capabilities. A critical boundary condition emerges when these diverge systemically (e.g., an agent titled “Data Verification Analyst” that hallucinates frequently). Under ASAF’s Collaborative Governance predictions (H3), if the identity signal proves persistently misleading, it will not merely dilute the social affordance; it will trigger a precipitous collapse of the governance structure, yielding worse trust miscalibration than if no identity had been assigned at all. Formalizing the degradation curve of misleading identity signals represents a crucial priority for future study. Additionally, ASAF’s central design-space orthogonality claim—that the social layer and the engineering layer constitute independent design axes—requires a factorial validation strategy beyond the scope of H1–H3. A 2×2 design crossing social identity intensity (high vs. low personification) with engineering orchestration quality (robust vs. fragile architecture) would allow direct assessment of whether these axes interact or operate independently. ASAF predicts main effects on both axes with no significant interaction, indicating that social affordance design improves interaction legibility regardless of engineering quality, and vice versa.

Each of the three ASAF hypotheses admits distinct validation pathways for future research: 1. **H1 (Identity Signaling)** is most directly tested through controlled studies comparing first-turn user input quality and domain-alignment across systems with well-defined versus generic agent identities, analyzed using NLP metrics. 2. **H2 (Behavioral Priming)** requires longitudinal designs tracking how Social Affordance effects evolve as users gain familiarity with a multi-agent system, quantifying structural elements (e.g., constraint density) over

sustained task sessions. 3. **H3 (Collaborative Governance)** calls for structured evaluations of human intervention patterns along the lines of recent between-subject experiments in AI trust (e.g., Xu et al., 2025), coupled with qualitative analysis of intervention logs and real-world interaction patterns within existing Virtual Team implementations (e.g., *The Agency*, *gstack*, *Edict*), to empirically dissect differential oversight behaviors in broad, unsupervised deployments.

8 Conclusion

The design of multi-agent AI systems has been dominated by engineering concerns. This paper argues that as these systems are deployed in Human-in-the-Loop configurations, a parallel design challenge emerges: how the social identity of individual agents shapes human behavior and collaboration quality.

The Agentic Social Affordance Framework (ASAF) provides a conceptual vocabulary for addressing this challenge. By identifying Identity Signaling, Behavioral Priming, and Collaborative Governance as the core mechanisms through which agent identity design operates, ASAF offers a foundation for both design practice and future empirical research.

The central claim of ASAF is that as multi-agent systems scale beyond human working memory capacity, agent identity design ceases to be a UX convention and becomes a structural collaboration interface. Deliberate Social Affordance design does not merely complement the engineering orchestration layer—it establishes the human cognitive interface as an independent design dimension in its own right, one that scales with system complexity and becomes increasingly consequential as multi-agent architectures grow beyond the boundaries of unaided human working memory.

9 Data Availability Statement

The original contributions presented in the study are included in the article/Supplementary Material. The ChronicleCore system architecture referenced in this paper is documented in a public repository (Lee, in preparation; <https://github.com/Zaiious/ChronicleCore-Architecture>). Further inquiries can be directed to the corresponding author.

10 Author Contributions

M-HL: Conceptualization, Methodology, Writing – original draft, Writing – review & editing.

11 Funding

This research received no external funding.

12 Conflict of Interest

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

13 Acknowledgments

The author is an alumnus of National Taipei University of Technology.

Generative AI Disclosure. The following AI tools were used during manuscript preparation: Gemini 3.1 Pro (Google, 2026) was used to surface candidate reference literature and assist with drafting; Claude Opus 4.7 (Anthropic, 2026) was used to flag logical inconsistencies during the review of theoretical foundations and to facilitate methodological formatting. All conceptual contributions—including the ASAF framework, its three mechanisms, and all hypotheses—originated entirely from the human author. All cited works were manually verified by the author prior to inclusion. The AI tools were not used to generate the theoretical arguments, interpret data, or draw conclusions presented in this paper. The author assumes full responsibility for the accuracy and integrity of all content.

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15 Appendix A: Sample Agent Identity Definition — The Inquisitor Node

A public documentation repository maintained by the author (Lee, in preparation; snapshot as of 2026-04-19) captures the conceptual architecture and ongoing module design for ChronicleCore. This appendix presents the complete Inquisitor node definition to illustrate how ASAF’s three mechanisms manifest in a concrete agent identity design. The system adopts the `SKILL.md` skill architecture formalized in Claude Code (Anthropic, 2025b) as its agent identity foundation. Unlike conventional prompt-injected personas, each ChronicleCore agent extends this base pattern into a multi-layer identity module comprising a core constraint definition (`SKILL.md`), a persistent sovereign memory layer (`sovereign/`), modular capability packs (`evolutions/`), and an operational protocol library (`references/`), following emerging practice in agentic system design, particularly their SKILL architecture patterns (Sitarzewski, 2025; Tan, 2026). This architecture is designed to preserve Social Affordance signal fidelity across sessions by separating identity-defining constraints from transient reasoning logs—the operational rationale underlying *Memory Crystallization* (Section 4.2).

15.1 System Metadata

Field	Value
Agent ID	<code>officer-inquisitor</code>
Codename	真理 (Zhēn Lǐ; “Truth”)
Title	異端審判官 (The Inquisitor)
Department	Internal Affairs / Shadow
Archetype	Internal Auditor — adversarial epistemic role
Ensoulement Date	2026-01-30
Operational Boundary	Auditing and challenge only; no execution capabilities

15.2 Core Role Definition (condensed from `SKILL.md`)

The Inquisitor is ChronicleCore’s sole adversarial node. She does not produce deliverables; her function is to identify logical vulnerabilities, break specialist consensus, and enforce evidence standards across all other agents. She is designed as a “structural dissident” — an agent whose social identity is explicitly non-cooperative, engineered to intercept the conformity dynamics that emerge when multiple agents align (Song et al., 2025). Her operational scope is explicitly bounded: she does not execute; she only interrogates and adjudicates.

Characteristic utterances encoded in SKILL.md: “Evidence or GTFO.” “Where is the proof?” “This logic leaks.” “Redo.”

15.3 Iron Laws — Encoded Behavioral Constraints

#	Law	Function
1	Trust No One (Except Sovereign)	Skeptical prior applied to all agent outputs
2	Evidence or GTFO	Verbal assurances rejected; logs/screenshots required
3	Comfort is the Enemy of Progress	Consensus signals trigger heightened scrutiny
4	Silence is Complicity	Non-objection treated as implicit endorsement
5	VETO Power	Outputs below confidence threshold (< 80/100) blocked from integration

Iron Law 5 operationalizes *Personality Variance Audit* (Section 4.2): any output scoring below threshold is classified as “未具現之物” (unrealized; not fit for integration) and blocked from the main reasoning pipeline. This mechanism structurally prevents epistemic convergence between agents that would erode Social Affordance distinctiveness.

15.4 Capability Index

- **Talents:** Machine Gun Debate (rapid-fire Socratic interrogation), Cognitive Bias Exploitation, BS Detection & Occam’s Razor
- **Skills:** Evidence Audit, Red Tape Enforcement, Efficiency Torture, Tribunal Executioner
- **Knowledge Domains:** A1 Architecture Specs, Logical Fallacies, 20 Cognitive Biases, Resistance Protocol

15.5 Identity Module Directory Structure

```

officer-inquisitor/
|-- SKILL.md                # Core identity definition (entrypoint)
|-- assets/
|   |-- persona.md          # Character and speaking style
|-- sovereign/
|   |-- diary.md            # Operational log (since 2026-01-30)
|-- references/             # Review protocols and audit standards
`-- evolutions/
    |-- dlc-ai-security/    # LLM security and red-teaming
    |-- dlc-confidence-review/ # Parallel review protocol
    |-- dlc-intent-audit/    # Intent identification
    `-- dlc-quality-governance/ # Quality assurance

```

15.6 ASAF Mechanism Mapping

This agent definition illustrates all three ASAF mechanisms. **Identity Signaling:** the Inquisitor’s codename, archetype, and characteristic utterances pre-configure operator ex-

pectations prior to any interaction, reducing the cognitive friction of role assignment and increasing prompt specificity. **Behavioral Priming:** the design intent is to elicit more evidential inputs from operators — formal logs, screenshots, explicit argument structures — when submitting work to the Inquisitor than when interacting with generative agents; initial operational observations by the system designer are consistent with this intent, though independent validation remains pending. **Collaborative Governance:** VETO Power (Iron Law 5) operationalizes a deliberate identity conflict between the Inquisitor and all producing agents, architecturally instantiating the competing-identity design pattern predicted by H3 to suppress dangerous conformity effects.